

SatQuery-Edge

Offline Satellite Intelligence Investigation Report
SIH 26167 · ISRO · Prototype · Offline Mode

Investigation ID	SQE-20260827-C2X07P
Timestamp	2026-08-27 23:08:31
Intent	bi_temporal_change
Tools Used	change, grounding, geospatial

Query

Identify newly flooded built-up areas and show the highest-confidence regions.

Final Answer

Candidate newly flooded built-up regions detected.

- 2 significant changed region(s) identified.
- Approximately 16.3% of scene shows change.
- Evidence Score: 80/100 — HIGH CONFIDENCE.

Detected change signatures are consistent with new water-like inundation over areas previously identified as built-up terrain. Highest-confidence regions are highlighted in the overlay and mapped.

NOTE: Results are demonstration-grade. Not a scientifically validated remote-sensing product.

Evidence Score

Component	Code	Value	Weight
Model Agreement	M	0.873	0.30
Sensor Reliability	S	0.900	0.20
Spatial Consistency	P	0.733	0.20
Temporal Consistency	T	0.940	0.15
Input Quality	I	0.478	0.15
FINAL SCORE		80.1 / 100	

Validation

Status: **PASS** ✓

- 2 image(s) detected — count OK.
- Image 1: Quality OK (mean=159, std=35.5).
- Image 2: Quality OK (mean=153, std=36.3).

- Image pair: identical dimensions — perfect alignment.

Detected Regions & Approximate Coordinates

#	Label	Confidence	Area (km²)	Demo Georeferenced
1	candidate changed region	0.85	18.493	Yes
2	candidate changed region	0.76	9.770	Yes
3	candidate flooded built-up are	0.97	128.691	Yes

Specialist Tool Outputs

CHANGE — SUCCESS — 2 region(s)

Detected 2 significant candidate changed region(s).

Largest region:

Approximate change area: 10.1% of image.

Confidence: 0.85

Center: (0.37, 0.66) (normalized)

Total changed pixels: ~16.3% of scene.

Interpretation:

Candidate newly inundated / possible flooded regions where spectral properties changed significantly between observations.

New water-like spectral signatures appear over

GROUNDING — SUCCESS — 1 region(s)

Grounding found 1 candidate region(s) matching 'flooded built-up areas'.

Highest confidence: 0.97.

GEOSPATIAL — SUCCESS — 0 region(s)

Geospatial conversion complete.

Converted 3 region(s) to geographic coordinates.

Total approximate area: 156.95 km².

■ Approximate demo georeferencing applied.

Demo center: (20.296°N, 85.824°E)

Coordinates are NOT derived from real GeoTIFF metadata.

Execution Trace

[23:08:31] QUERY RECEIVED: Identify newly flooded built-up areas and show the highest-confidence regions.

[23:08:31] ROUTER DECISION: Intent: bi_temporal_change

Selected tools: change, grounding, geospatial

Images required: 2

[23:08:31] VALIDATION: Status: PASS

2 image(s) detected — count OK.

Image 1: Quality OK (mean=159, std=35.5).

Image 2: Quality OK (mean=153, std=36.3).

[23:08:31] TOOL EXECUTING: change → started

[23:08:31] TOOL OUTPUT (change): Success — 2 region(s) detected

Detected 2 significant candidate changed region(s).

Largest region:

[23:08:31] TOOL EXECUTING: grounding → started

[23:08:32] TOOL OUTPUT (grounding): Success — 1 region(s) detected

Grounding found 1 candidate region(s) matching 'flooded built-up areas'.

Highest confidence: 0.97.

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[23:08:32] TOOL EXECUTING: geospatial → started
[23:08:32] TOOL OUTPUT (geospatial): Success – 0 region(s) detected
Geospatial conversion complete.
Converted 3 region(s) to geographic coordinates.
Total approximate area: 156.95 km².
[23:08:32] EVIDENCE FUSION: Score: 80.1/100
M=0.873
S=0.900
P=0.733
[23:08:32] FINAL SCORE: 80/100
[23:08:32] ANSWER: Candidate newly flooded built-up regions detected.
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- 2 significant changed region(s) identified.
- Approximately 16.3%

Tool Versions

Component	Version
SatQuery-Edge Prototype	1.0.0-SIH26167
Python OpenCV	4.11.0
NumPy	1.26.4
Streamlit	1.62.0
Report Engine	ReportLab PDF
Mode	OFFLINE — No external APIs

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. — SIH 26167 · ISRO · SatQuery-Edge